Homework 3 Solutions

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Section 1.4

1.4–2. Let $P(A) = 0.3$, $P(B) = 0.6$.

(a) If A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B).$$

Substituting values,

$$0.3 + 0.6 - (0.3)(0.6) = 0.9 - 0.18 = 0.72.$$

(b) If A and B are mutually exclusive, then $P(A \cap B) = 0$. So

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{0.6} = 0.$$

1.4–5. Given $P(A) = 0.8$, $P(B) = 0.5$, $P(A \cup B) = 0.9$. First find

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.8 + 0.5 - 0.9 = 0.4.$$

For independence, we need $P(A \cap B) = P(A)P(B)$. Indeed,

$$P(A)P(B) = (0.8)(0.5) = 0.4,$$

so A and B are independent.

1.4–7. Probabilities: $P(A_1) = 0.5$, $P(A_2) = 0.7$, $P(A_3) = 0.6$.

(a) Probability that exactly one is successful:

$$0.5(1 - 0.7)(1 - 0.6) + 0.7(1 - 0.5)(1 - 0.6) + 0.6(1 - 0.5)(1 - 0.7).$$

$$= 0.5(0.3)(0.4) + 0.7(0.5)(0.4) + 0.6(0.5)(0.3) = 0.06 + 0.14 + 0.09 = 0.29.$$

(b) Probability exactly two succeed:

$$0.5(0.7)(1 - 0.6) + 0.5(0.6)(1 - 0.7) + (1 - 0.5)(0.7)(0.6).$$

$$= 0.35(0.4) + 0.30(0.3) + 0.50(0.42) = 0.14 + 0.09 + 0.21 = 0.44.$$

1.4–11.

(a) If A and B are mutually exclusive, then $P(A \cap B) = 0$. For independence we would need $P(A \cap B) = P(A)P(B)$. This can only happen if $P(A) = 0$ or $P(B) = 0$. So in general, mutually exclusive events are not independent (unless one is impossible).

(b) If $A \subseteq B$, then $P(A \cap B) = P(A)$. Independence would require $P(A) = P(A)P(B)$. So either $P(A) = 0$ or $P(B) = 1$. Only in those trivial cases are A and B independent.

1.4–12. A fair coin is flipped 5 times. Each outcome has probability $(1/2)^5 = 1/32$.

(a) $P(\text{HHTHT}) = 1/32$. (b) $P(\text{THHHT}) = 1/32$. (c) $P(\text{HTHTH}) = 1/32$. (d) Exactly 3 heads: there are $\binom{5}{3} = 10$ outcomes with 3 heads, so

$$P(3 \text{ heads}) = \frac{10}{32} = \frac{5}{16} \approx 0.3125.$$

Section 1.5

1.5–2. Supplier A provides 40% of the seeds with 85% germination, and Supplier B provides 60% with 75% germination.

(a) By the law of total probability:

$$P(G) = (0.4)(0.85) + (0.6)(0.75) = 0.34 + 0.45 = 0.79.$$

(b) By Bayes' rule:

$$P(A|G) = \frac{P(G|A)P(A)}{P(G)} = \frac{(0.85)(0.4)}{0.79} = \frac{0.34}{0.79} \approx 0.4304.$$

1.5–3. We are told:

$$P(\text{high}) = 0.16, \quad P(\text{low}) = 0.19, \quad P(\text{normal}) = 1 - 0.16 - 0.19 = 0.65.$$

$$P(\text{irregular}) = 0.17.$$

Also,

$$P(\text{high} \cap \text{irregular}) = (0.35)(0.17) = 0.0595,$$

$$P(\text{normal} \cap \text{irregular}) = (0.11)(0.65) = 0.0715.$$

So

$$P(\text{low} \cap \text{irregular}) = 0.17 - (0.0595 + 0.0715) = 0.039.$$

Thus the proportion with low blood pressure and a regular heartbeat is

$$P(\text{low}) - P(\text{low} \cap \text{irregular}) = 0.19 - 0.039 = 0.151 = 15.1\%.$$

1.5–4. We use the law of total probability:

$$P(\text{accident}) = (0.10)(0.05) + (0.55)(0.02) + (0.20)(0.03) + (0.15)(0.04).$$

$$= 0.005 + 0.011 + 0.006 + 0.006 = 0.028.$$

Now,

$$P(\text{16–25} \mid \text{accident}) = \frac{0.10 \cdot 0.05}{0.028} = \frac{0.005}{0.028} \approx 0.1786 \ (\approx 17.9\%).$$

1.5–8. Market shares:

$$P(B_1) = 0.40,\ P(B_2) = 0.30,\ P(B_3) = 0.20,\ P(B_4) = 0.10.$$

Repair probabilities: $0.10, 0.05, 0.03, 0.02$ respectively.

Total repair probability:

$$P(R) = (0.4)(0.10) + (0.3)(0.05) + (0.2)(0.03) + (0.1)(0.02) = 0.063.$$

Using Bayes' rule:

$$P(B_1|R) = \frac{(0.4)(0.10)}{0.063} = \frac{0.04}{0.063} \approx 0.635,$$

$$P(B_2|R) = \frac{0.015}{0.063} \approx 0.238, \quad P(B_3|R) = \frac{0.006}{0.063} \approx 0.095,$$

$$P(B_4|R) = \frac{0.002}{0.063} \approx 0.032.$$